

BHUVANESAN N

Software Developer | Java | Frontend | IoT

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PROFILE SUMMARY

Software Developer with hands-on experience across frontend and backend development, embedded systems, and applied machine learning. Built responsive web interfaces with HTML, CSS, JavaScript, and Bootstrap, backend CRUD applications with Java, Spring Boot, and MongoDB, and IoT prototypes using ESP32-based sensor systems. Currently expanding into AI/ML and Snowflake for data-driven and predictive applications. Comfortable working across the stack, from UI logic to database design to hardware-sensor integration. Actively strengthening problem-solving skills through DSA practice (150+ LeetCode problems solved) and looking to contribute to full-stack, IoT, and ML-driven engineering projects.

EDUCATION

B.E. Computer Science and Engineering (CSE) — PSNA College of Engineering and Technology, Dindigul

2024 – 2028 | CGPA: 8.6

XII — St. John's Matriculation Higher Secondary School, Madurai | May 2024 | 93%

X — The Nallamani Matriculation Higher Secondary School, Madurai | July 2022 | 92%

PROJECTS

Digital AI Twin

IoT · Python · AI/ML

- Built a digital twin platform that simulates physical systems using real-time IoT sensor data.
- Integrated prediction engines for traffic, energy consumption, and environmental monitoring.
- Enabled data-driven analysis and intelligent decision support through virtual system modeling.

Smart Energy Meter

ESP32 · LCD · Embedded Systems · IoT

- Built a real-time energy monitoring system using ESP32, an LCD display, and a potentiometer to simulate live current measurement.
- Calculated and displayed Current (A), Power (W), and Energy Consumption (kWh) in real time from sensor input.
- Designed the system to demonstrate smart metering concepts in line with SDG 7 – Affordable and Clean Energy, supporting electricity usage monitoring and energy efficiency.

Employee Management System

HTML · CSS · SQL

- Built an Employee Management System with an HTML/CSS frontend and a SQL backend for storing and managing employee records.
- Implemented core CRUD operations (add, update, delete, view employee data) using SQL queries connected to the frontend interface.
- Designed simple, structured web forms for data entry and retrieval of employee information.

Micro Failure Whisperer (Ongoing)

Python · Machine Learning · Predictive Maintenance

- Creating an industrial machine diagnosis system that detects micro-level failures from sensor and vibration data.
- Applying anomaly detection and predictive maintenance techniques to flag faults before breakdowns occur.
- Designing a real-time monitoring pipeline for industrial environments.

Digital Smell (Ongoing)

IoT · Gas Sensors · Embedded AI

- Designing a system to detect minute gas leakages using high-sensitivity gas sensors and embedded AI models.
- Implementing real-time alerting and anomaly detection for safety-critical environments.

ScriptGuard AI (Ongoing)

Python · Computer Vision · OCR

- Developing an intelligent paper-scanning mechanism for automated document digitization and verification.
- Integrating OCR and image-processing techniques to extract, validate, and organize scanned content.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++

Frontend & Web Technologies: HTML5, CSS3, JavaScript, Bootstrap

Backend & Databases: Core Java, Spring Boot, MongoDB, MySQL (SQL), CRUD Operations
AI/ML: Machine Learning Fundamentals, Anomaly Detection, Predictive Maintenance
Cloud & Data: Snowflake
IoT & Embedded: ESP32, Sensor Integration, Embedded AI, Gas & Vibration Sensors
Tools & Platforms: VS Code, IntelliJ IDEA, Eclipse, Git, GitHub
Concepts: OOP, Data Structures & Algorithms, DOM Manipulation, Responsive Web Design, Anomaly Detection

INTERPERSONAL SKILLS

- | | | |
|-----------------|-------------------|-----------------------|
| ● Communication | ● Problem-Solving | ● Time Management |
| ● Presentation | ● Team Player | ● Attention to Detail |
| ● Adaptability | | |

WORKSHOPS & HACKATHONS

HACKXTREME'26 — 24-Hour National-Level Hackathon, K. Ramakrishnan College of Engineering, Trichy

- Built an AI-powered Digital Twin model simulating urban infrastructure from IoT data for real-time prediction of traffic congestion, energy demand, and pollution levels.
- Recognized by Media Wave for technical performance and shortlisted for an internship opportunity.

Smart India Hackathon (SIH)

- Designed an automated pesticide sprinkler system with logic-based control to minimize manual intervention and improve spraying efficiency at the national level.

MSME

- Presented Micro Failure Whisperer and Digital Smell as MSME-aligned solutions for industrial safety and predictive maintenance.

Snowflake Workshop — 1 Week

- Completed a week-long hands-on workshop covering Snowflake fundamentals for cloud data warehousing.

IoT Workshop — 1 Month

- Completed a month-long IoT workshop covering sensor integration, embedded systems, and real-time data monitoring.

CERTIFICATIONS

- | | | |
|---------------------|-----------------------|---|
| ● AI/ML — Dreams AI | ● SQL — Oracle | ● Java — Oracle |
| ● MongoDB — ICT | ● Figma — Simplilearn | ● Python, C & C++ —
Kaggle/GreatLearning |

ACHIEVEMENTS

- Recognized by Media Wave at HACKXTREME'26 (24-Hour National-Level Hackathon) for technical performance and shortlisted for an internship opportunity.
- Solved 170+ problems on LeetCode across Arrays, Strings, Mathematics, and core DSA topics.
- Won at math events, enhancing quick and logical thinking and reasoning skills.